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NDIC - DATEX II Situation Publication - Parking occupancy

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The document describes a data format used for publishing parking occupancy.

CHAPTER 1

Introduction

This document shall be used by publisher and / or subscriber. Subscriber shall use it for format suitability evaluation and for possible implementation of received NTIC data import into own information system. Publisher shall use it as a specification of data to be published format and for possible implementation of data export from own information system into NTIC.

Here you can find:

- conceptual description of published information
- information about location referencing methods used
- W3C XML schema of the format
- data sample(s)

Following topics are out of scope of this document and if present, shall be considered as informative only:

- transfer protocol incl. access point, life cycle etc.
- related data formats
- typical data origin
- typical data use

1.1 General terms

Following terms are used in this document:

format description of data structure for transferring some kind of information

protocol description of data transfer process and rules

NDIC national traffic information centre

DATEX II model class model for describing information related to the road traffic. The model in form of UML is platform independent.

One of specific platforms for expressing the model is XML.

The model is extensive (it consists of few thousands of structural items representing term incl. definition) For reference data (UML class model in form of Enterprise Architect EAP file, XML schema etc.) see *[d2ref]*. For additional information see *[d2supp]*.

(DATEX II) profile simplified class model, still sufficient for specific needs. Profile is created from model by selecting only specific parts.

(DATEX II) extension extended DATEX II model, adding data structures missing in standard model but needed for specific needs.

B level extension Extension of DATEX II model, which is backward compatible with the original DATEX II model, even though it contains additional structures. *[16157-1]*

C level extension Extension of DATEX II model, which is not backward compatible with the original DATEX II model.

publication an element of DATEX II model, serving as an envelope for transmitted content of certain kind. Every sent DATEX II message contains exactly one element of publication type. There are various types of publications, for example *SituationPublication [16157-3]*, *ElaboratedDataPublication [16157-5]*, *PredefinedLocationsPublication [16157-3]* etc.

(W3C) XML schema language, which allows definition of rules for a XML document structure.

XSD XML document containing XML schema. Sometimes it is synonym for XML schema itself.

uri uniform resource identifier, text, used as unique identifier. In this document, every format has it's own uri.

CHAPTER 2

Concepts

Described format allows publishing parking occupancy related information for a set of predefined locations. This information belongs to category of traffic data (status), it is published in predefined intervals, the contents carry percentages of occupancy.

2.1 Publication type SituationPublication

Format is based on DATEX II publication of type *SituationPublication* *[16157-3]*. Profile uses only small subset of possible structures.

```
<?xml version="1.0" encoding="utf-8"?>
<d2LogicalModel modelBaseVersion="2" xmlns="http://datex2.eu/schema/2/2_0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
  <exchange>
    <supplierIdentification>
      <country>cz</country>
      <nationalIdentifier>ITSPARKING</nationalIdentifier>
    </supplierIdentification>
  </exchange>
  <payloadPublication lang="cs" xsi:type="SituationPublication">
    <feedType>x-format:cz-ndic_d2-common-v1.1</feedType>
    <publicationTime>2016-05-18T18:49:21+02:00</publicationTime>
    <publicationCreator>
      <country>cz</country>
      <nationalIdentifier>DRU</nationalIdentifier>
    </publicationCreator>
```

2.2 Event description (only NonRoadEventInformation/CarParks)

SituationPublication requires description by means of class *Situation*, which aggregates instances of class *SituationRecord*, namely descendants of class *CarParks*. Besides the *SituationRecord* it contains elements with information about service availability.

Element *situation* describes one occupancy information, more particular occupancy information means more *situation* elements. Since the situation described may evolve over time, the element is versioned and has unique identifier. Each *situation* element contains one *situationRecord* element.

The *situationRecord* element may, in the element *situation*, occur exactly once. The *situationRecord* element may evolve over time, so the element is versioned and has unique identifier.

```xml
    <situation id="PR-2" version="8387">
      <situationVersionTime>
      2016-05-18T18:48:51+02:00</situationVersionTime>
      <headerInformation>
        <confidentiality>noRestriction</confidentiality>
        <informationStatus>real</informationStatus>
        <urgency>normalUrgency</urgency>
      </headerInformation>
      <situationRecord id="PR-2_CarParks" version="8387"
      xsi:type="CarParks">
        <situationRecordCreationTime>
        2016-05-18T18:49:21+02:00</situationRecordCreationTime>
        <situationRecordVersionTime>
        2016-05-18T18:48:51+02:00</situationRecordVersionTime>
        <probabilityOfOccurrence>riskOf</probabilityOfOccurrence>
        <source>
          <sourceIdentification>ITSPARKING</sourceIdentification>
          <sourceName>
            <values>
              <value lang="cs">Správci parkovišť</value>
            </values>
          </sourceName>
        </source>
        <validity>
          <validityStatus>definedByValidityTimeSpec</validityStatus>
          <validityTimeSpecification>
            <overallStartTime>2016-05-18T18:48:51+02:00</overallStartTime>
            <overallEndTime>2016-05-18T19:03:51+02:00</overallEndTime>
          </validityTimeSpecification>
        </validity>
        <generalPublicComment>
          <comment>
            <values>
              <value lang="cs">P+R Černý Most, Chlumecká, 50% obsazeno,
              18.05.2016 18:48:51</value>
            </values>
          </comment>
        </generalPublicComment>
        <groupOfLocations xsi:type="LocationByReference">
          <predefinedLocationReference id="98D69F80-7E63-422B-A9B7-EEACE24C055C"
          targetClass="PredefinedLocation" version="1" />
        </groupOfLocations>
        <carParkIdentity>P+R Černý Most</carParkIdentity>
        <carParkOccupancy>50</carParkOccupancy>
      </situationRecord>
```

2.3 Location description (by predefined locations)

The position can be described by a point or a short linear section on the road network, given by the presence of a car park. Since the position of the car parks does not change, the location in this profile is described by a reference (id) to a predefined location set defined in different document. The reference points to a class instance *Predefined-Location* from package *PredefinedLocationsPublication*. Therefore, the location is not described directly in the file.

Location of a car park is determined by reference to a set of predefined locations. To look up given location, it is necessary to obtain publication *PredefinedLocationsPublication*, related to this source of information and inside find location of identical type (by *targetClass*, here *PredefinedLocation*), with the same identifier and version.

The location for which the parking occupancy information is valid is specified by the *groupOfLocations* element re-typed to *xsi:type="LocationByReference"*, that includes element *predefinedLocationReference* with the attribute *targetClass="PredefinedLocation"*.

```
        <groupOfLocations xsi:type="LocationByReference">
          <predefinedLocationReference id="98D69F80-7E63-422B-A9B7-EEACE24C055C"
          targetClass="PredefinedLocation" version="1" />
        </groupOfLocations>
```

CHAPTER 3

Time validity

Time validity is stated independently for every single *SituationRecord* described in a situation and is defined by a time interval between two time instants. The class *Validity* is used with an instance of class *OverallPeriod* in the role of *validityTimeSpecification* and elements *overallStartTime* and *overallStopTime* to define start and end of the interval.

Apart from methods mentioned above, no other methods for time validity specification are allowed (e.g. hours of validity or days of validity).

```
        <validity>
          <validityStatus>definedByValidityTimeSpec</validityStatus>
          <validityTimeSpecification>
            <overallStartTime>2016-05-18T18:48:51+02:00</overallStartTime>
            <overallEndTime>2016-05-18T19:03:51+02:00</overallEndTime>
          </validityTimeSpecification>
        </validity>
```

CHAPTER 4

Location referencing

The format describes routes exclusively by a reference to set of predefined locations (class *PredefinedLocation*) in publication *PredefinedLocationsPublication* [16157-3].

Location referencing methods are out of scope of this document and are described elsewhere.

CHAPTER 5

Appendix A: Message examples

5.1 parking.xml

Publication contains exactly one message about one car park occupied at 50 %.

5.1.1 Root element and publication envelope

Root element *d2LogicalModel* declares, that this message is using DATEX II.

Element *exchange* contains provider identifier.

Element *payloadPublication* is an envelope for actual information payload containing general information and one or more elements *situation*, (in our case there is one).

Attribute *xsi:type="SituationPublication"* in element *payloadPublication* determines, that this envelope is of type Situation publication.

Element *feedType* states publication schema by its uri.

Element *publicationTime* states publication time of this publication (file) as a whole.

Element *publicationCreator* identifies creator of the publication.

5.1.2 Description of one specific situation

Element *situation* describes a particular event, more particular events means more *situation* elements. Each *situation* element may contain one element *situationRecord*.

Element *situationVersionTime* sets the time of the publication of this version of the situation.

Element *headerInformation* describes how to handle the information described.

Event description (what happened)

The attribute *xsi:type="CarParks"* of element *situationRecord* defines the event type. In DATEX II model *[d2ref]* it is only used class in this profile.

From this class, elements *carParkIdentity* are used with value of *P+R Černý Most* and element *carParkOccupancy* specifies the occupancy being *50* percent.

Time validity

Estimated time validity is given in a structure *validity* by values of validity interval start *overallStartTime* and validity interval end *overallEndTime*.

Event location

Event location is determined by reference to a set of predefined locations. To look up given location, it is necessary to obtain publication *PredefinedLocationsPublication*, related to this source of information and inside find location of identical type (by *targetClass*, here *PredefinedLocation*), with the same identifier and version.

The location for which the parking occupancy is valid is specified by the *groupOfLocations* element. It always has *xsi:type="LocationByReference"* and includes element *predefinedLocationReference* with the attribute *targetClass="PredefinedLocation"*.

The attribute *id ="98D69F80-7E63-422B-A9B7-EEACE24C055C"* refers to a predefined location with a given identifier, and the attribute *version ='1'* specifies the version of the location description.

Additional content

The document states time of its first version in *situationRecordCreationTime* and time of current version in *situationRecordVersionTime* for each *situationRecord*.

Element *source* lists the the creator of this information, who is responsible for its accuracy.

Element *generalPublicComment* contains a general description of the situation, mostly only in Czech language. This is a free text description as provided by the situation creator.

CHAPTER 6

Appendix B: W3C XML Schema

XML structure of a transmitted message is defined by following W3C XML schema:

6.1 *schema.xsd*

```
<?xml version="1.0" encoding="utf-8" standalone="no"?>
<xs:schema elementFormDefault="qualified" attributeFormDefault="unqualified" xmlns:D2LogicalModel="http://datex2.eu/schema/2/2_0" version="2.3" targetNamespace="http://datex2.eu/schema/2/2_0" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:complexType name="_ExtensionType">
    <xs:sequence>
      <xs:any namespace="##any" processContents="lax" minOccurs="0" maxOccurs="unbounded" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="_PredefinedLocationVersionedReference">
    <xs:complexContent>
      <xs:extension base="D2LogicalModel:VersionedReference">
        <xs:attribute name="targetClass" use="required" fixed="PredefinedLocation" />
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="AxleFlowValue">
    <xs:complexContent>
      <xs:extension base="D2LogicalModel:DataValue">
        <xs:sequence>
          <xs:element name="axleFlowRate" type="D2LogicalModel:AxlesPerHour" minOccurs="1" maxOccurs="1" />
          <xs:element name="axleFlowValueExtension" type="D2LogicalModel:_ExtensionType" minOccurs="0" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:simpleType name="AxlesPerHour">
```

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```
    <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
  </xs:simpleType>
  <xs:simpleType name="Boolean">
    <xs:restriction base="xs:boolean" />
  </xs:simpleType>
  <xs:complexType name="CarParks">
    <xs:complexContent>
      <xs:extension base="D2LogicalModel:NonRoadEventInformation">
        <xs:sequence>
          <xs:element name="carParkIdentity" type="D2LogicalModel:String"␣
↪minOccurs="1" maxOccurs="1" />
          <xs:element name="carParkOccupancy" type="D2LogicalModel:Percentage"␣
↪minOccurs="0" maxOccurs="1" />
          <xs:element name="carParksExtension" type="D2LogicalModel:_ExtensionType
↪" minOccurs="0" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="Comment">
    <xs:sequence>
      <xs:element name="comment" type="D2LogicalModel:MultilingualString"␣
↪minOccurs="1" maxOccurs="1" />
      <xs:element name="commentDateTime" type="D2LogicalModel:DateTime" minOccurs=
↪"0" maxOccurs="1" />
      <xs:element name="commentType" type="D2LogicalModel:CommentTypeEnum"␣
↪minOccurs="0" maxOccurs="1" />
      <xs:element name="commentExtension" type="D2LogicalModel:_ExtensionType"␣
↪minOccurs="0" />
    </xs:sequence>
  </xs:complexType>
  <xs:simpleType name="CommentTypeEnum">
    <xs:restriction base="xs:string">
      <xs:enumeration value="abnormalLoadMovementNote" />
      <xs:enumeration value="dataProcessingNote" />
      <xs:enumeration value="description" />
      <xs:enumeration value="internalNote" />
      <xs:enumeration value="locationDescriptor" />
      <xs:enumeration value="warning" />
      <xs:enumeration value="other" />
    </xs:restriction>
  </xs:simpleType>
  <xs:simpleType name="ComputationMethodEnum">
    <xs:restriction base="xs:string">
      <xs:enumeration value="arithmeticAverageOfSamplesBasedOnAFixedNumberOfSamples
↪" />
      <xs:enumeration value="arithmeticAverageOfSamplesInATimePeriod" />
      <xs:enumeration value="harmonicAverageOfSamplesInATimePeriod" />
      <xs:enumeration value="medianOfSamplesInATimePeriod" />
      <xs:enumeration value="movingAverageOfSamples" />
    </xs:restriction>
  </xs:simpleType>
  <xs:complexType name="ConcentrationOfVehiclesValue">
    <xs:complexContent>
      <xs:extension base="D2LogicalModel:DataValue">
        <xs:sequence>
          <xs:element name="concentrationOfVehicles" type=
↪"D2LogicalModel:ConcentrationVehiclesPerKilometre" minOccurs="1" maxOccurs="1" />
          <xs:element name="concentrationOfVehiclesValueExtension" type=
↪"D2LogicalModel:_ExtensionType" minOccurs="0" />
        </xs:sequence>
```

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```
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:simpleType name="ConcentrationVehiclesPerKilometre">
    <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
  </xs:simpleType>
  <xs:simpleType name="ConfidentialityValueEnum">
    <xs:restriction base="xs:string">
      <xs:enumeration value="internalUse" />
      <xs:enumeration value="noRestriction" />
      <xs:enumeration value="restrictedToAuthorities" />
      <xs:enumeration value="restrictedToAuthoritiesAndTrafficOperators" />
      <xs:enumeration value="restrictedToAuthoritiesTrafficOperatorsAndPublishers"␣
↪/>
      <xs:enumeration value="restrictedToAuthoritiesTrafficOperatorsAndVms" />
    </xs:restriction>
  </xs:simpleType>
  <xs:simpleType name="CountryEnum">
    <xs:restriction base="xs:string">
      <xs:enumeration value="at" />
      <xs:enumeration value="be" />
      <xs:enumeration value="bg" />
      <xs:enumeration value="ch" />
      <xs:enumeration value="cs" />
      <xs:enumeration value="cy" />
      <xs:enumeration value="cz" />
      <xs:enumeration value="de" />
      <xs:enumeration value="dk" />
      <xs:enumeration value="ee" />
      <xs:enumeration value="es" />
      <xs:enumeration value="fi" />
      <xs:enumeration value="fo" />
      <xs:enumeration value="fr" />
      <xs:enumeration value="gb" />
      <xs:enumeration value="gg" />
      <xs:enumeration value="gi" />
      <xs:enumeration value="gr" />
      <xs:enumeration value="hr" />
      <xs:enumeration value="hu" />
      <xs:enumeration value="ie" />
      <xs:enumeration value="im" />
      <xs:enumeration value="is" />
      <xs:enumeration value="it" />
      <xs:enumeration value="je" />
      <xs:enumeration value="li" />
      <xs:enumeration value="lt" />
      <xs:enumeration value="lu" />
      <xs:enumeration value="lv" />
      <xs:enumeration value="ma" />
      <xs:enumeration value="mc" />
      <xs:enumeration value="mk" />
      <xs:enumeration value="mt" />
      <xs:enumeration value="nl" />
      <xs:enumeration value="no" />
      <xs:enumeration value="pl" />
      <xs:enumeration value="pt" />
      <xs:enumeration value="ro" />
      <xs:enumeration value="se" />
      <xs:enumeration value="si" />
      <xs:enumeration value="sk" />
      <xs:enumeration value="sm" />
```

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```
      <xs:enumeration value="tr" />
      <xs:enumeration value="va" />
      <xs:enumeration value="other" />
    </xs:restriction>
  </xs:simpleType>
  <xs:element name="d2LogicalModel" type="D2LogicalModel:D2LogicalModel">
    <xs:unique name="_d2LogicalModelSituationRecordConstraint">
      <xs:selector xpath=".//D2LogicalModel:situationRecord" />
      <xs:field xpath="@id" />
      <xs:field xpath="@version" />
    </xs:unique>
    <xs:unique name="_d2LogicalModelSituationConstraint">
      <xs:selector xpath=".//D2LogicalModel:situation" />
      <xs:field xpath="@id" />
      <xs:field xpath="@version" />
    </xs:unique>
  </xs:element>
  <xs:complexType name="D2LogicalModel">
    <xs:sequence>
      <xs:element name="exchange" type="D2LogicalModel:Exchange" />
      <xs:element name="payloadPublication" type="D2LogicalModel:PayloadPublication
↪" minOccurs="0" />
      <xs:element name="d2LogicalModelExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
    </xs:sequence>
    <xs:attribute name="modelBaseVersion" use="required" fixed="2" />
  </xs:complexType>
  <xs:complexType name="DataValue" abstract="true">
    <xs:sequence>
      <xs:element name="dataError" type="D2LogicalModel:Boolean" minOccurs="0"␣
↪maxOccurs="1" />
      <xs:element name="reasonForDataError" type="D2LogicalModel:MultilingualString
↪" minOccurs="0" maxOccurs="1" />
      <xs:element name="dataValueExtension" type="D2LogicalModel:_ExtensionType"␣
↪minOccurs="0" />
    </xs:sequence>
    <xs:attribute name="accuracy" type="D2LogicalModel:Percentage" use="optional" /
↪>
    <xs:attribute name="computationalMethod" type=
↪"D2LogicalModel:ComputationMethodEnum" use="optional" />
    <xs:attribute name="numberOfIncompleteInputs" type=
↪"D2LogicalModel:NonNegativeInteger" use="optional" />
    <xs:attribute name="numberOfInputValuesUsed" type=
↪"D2LogicalModel:NonNegativeInteger" use="optional" />
    <xs:attribute name="smoothingFactor" type="D2LogicalModel:Float" use="optional
↪" />
    <xs:attribute name="standardDeviation" type="D2LogicalModel:Float" use=
↪"optional" />
    <xs:attribute name="supplierCalculatedDataQuality" type=
↪"D2LogicalModel:Percentage" use="optional" />
  </xs:complexType>
  <xs:simpleType name="DateTime">
    <xs:restriction base="xs:dateTime" />
  </xs:simpleType>
  <xs:complexType name="DateTimeValue">
    <xs:complexContent>
      <xs:extension base="D2LogicalModel:DataValue">
        <xs:sequence>
          <xs:element name="dateTime" type="D2LogicalModel:DateTime" minOccurs="1"␣
↪maxOccurs="1" />
          <xs:element name="dateTimeValueExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
```

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```
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="ElaboratedDataFault">
    <xs:complexContent>
      <xs:extension base="D2LogicalModel:Fault">
        <xs:sequence>
          <xs:element name="elaboratedDataFault" type=
↪"D2LogicalModel:ElaboratedDataFaultEnum" minOccurs="1" maxOccurs="1" />
          <xs:element name="elaboratedDataFaultExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:simpleType name="ElaboratedDataFaultEnum">
    <xs:restriction base="xs:string">
      <xs:enumeration value="intermittentDataValues" />
      <xs:enumeration value="noDataValuesAvailable" />
      <xs:enumeration value="spuriousUnreliableDataValues" />
      <xs:enumeration value="unspecifiedOrUnknownFault" />
      <xs:enumeration value="other" />
    </xs:restriction>
  </xs:simpleType>
  <xs:complexType name="Exchange">
    <xs:sequence>
      <xs:element name="supplierIdentification" type=
↪"D2LogicalModel:InternationalIdentifier" />
      <xs:element name="exchangeExtension" type="D2LogicalModel:_ExtensionType"␣
↪minOccurs="0" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="Fault">
    <xs:sequence>
      <xs:element name="faultIdentifier" type="D2LogicalModel:String" minOccurs="0
↪" maxOccurs="1" />
      <xs:element name="faultDescription" type="D2LogicalModel:String" minOccurs="0
↪" maxOccurs="1" />
      <xs:element name="faultCreationTime" type="D2LogicalModel:DateTime"␣
↪minOccurs="0" maxOccurs="1" />
      <xs:element name="faultLastUpdateTime" type="D2LogicalModel:DateTime"␣
↪minOccurs="1" maxOccurs="1" />
      <xs:element name="faultSeverity" type="D2LogicalModel:FaultSeverityEnum"␣
↪minOccurs="0" maxOccurs="1" />
      <xs:element name="faultExtension" type="D2LogicalModel:_ExtensionType"␣
↪minOccurs="0" />
    </xs:sequence>
  </xs:complexType>
  <xs:simpleType name="FaultSeverityEnum">
    <xs:restriction base="xs:string">
      <xs:enumeration value="low" />
      <xs:enumeration value="medium" />
      <xs:enumeration value="high" />
      <xs:enumeration value="unknown" />
    </xs:restriction>
  </xs:simpleType>
  <xs:simpleType name="Float">
    <xs:restriction base="xs:float" />
  </xs:simpleType>
  <xs:complexType name="GroupOfLocations" abstract="true">
```

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```
    <xs:sequence>
      <xs:element name="groupOfLocationsExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="HeaderInformation">
    <xs:sequence>
      <xs:element name="confidentiality" type=
↪"D2LogicalModel:ConfidentialityValueEnum" minOccurs="1" maxOccurs="1" />
      <xs:element name="informationStatus" type=
↪"D2LogicalModel:InformationStatusEnum" minOccurs="1" maxOccurs="1" />
      <xs:element name="urgency" type="D2LogicalModel:UrgencyEnum" minOccurs="0"␣
↪maxOccurs="1" />
      <xs:element name="headerInformationExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
    </xs:sequence>
  </xs:complexType>
  <xs:simpleType name="InformationStatusEnum">
    <xs:restriction base="xs:string">
      <xs:enumeration value="real" />
      <xs:enumeration value="securityExercise" />
      <xs:enumeration value="technicalExercise" />
      <xs:enumeration value="test" />
    </xs:restriction>
  </xs:simpleType>
  <xs:simpleType name="Integer">
    <xs:restriction base="xs:integer" />
  </xs:simpleType>
  <xs:complexType name="InternationalIdentifier">
    <xs:sequence>
      <xs:element name="country" type="D2LogicalModel:CountryEnum" minOccurs="1"␣
↪maxOccurs="1" />
      <xs:element name="nationalIdentifier" type="D2LogicalModel:String" minOccurs=
↪"1" maxOccurs="1" />
      <xs:element name="internationalIdentifierExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
    </xs:sequence>
  </xs:complexType>
  <xs:simpleType name="Language">
    <xs:restriction base="xs:language" />
  </xs:simpleType>
  <xs:complexType name="Location" abstract="true">
    <xs:complexContent>
      <xs:extension base="D2LogicalModel:GroupOfLocations">
        <xs:sequence>
          <xs:element name="locationExtension" type="D2LogicalModel:_ExtensionType
↪" minOccurs="0" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="LocationByReference">
    <xs:complexContent>
      <xs:extension base="D2LogicalModel:Location">
        <xs:sequence>
          <xs:element name="predefinedLocationReference" type="D2LogicalModel:_
↪PredefinedLocationVersionedReference" minOccurs="1" maxOccurs="1" />
          <xs:element name="locationByReferenceExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
        </xs:sequence>
      </xs:extension>
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```
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="MeasurementEquipmentFault">
    <xs:complexContent>
      <xs:extension base="D2LogicalModel:Fault">
        <xs:sequence>
          <xs:element name="measurementEquipmentFault" type=
↪"D2LogicalModel:MeasurementEquipmentFaultEnum" minOccurs="1" maxOccurs="1" />
          <xs:element name="measurementEquipmentFaultExtension" type=
↪"D2LogicalModel:_ExtensionType" minOccurs="0" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:simpleType name="MeasurementEquipmentFaultEnum">
    <xs:restriction base="xs:string">
      <xs:enumeration value="intermittentDataValues" />
      <xs:enumeration value="noDataValuesAvailable" />
      <xs:enumeration value="spuriousUnreliableDataValues" />
      <xs:enumeration value="unspecifiedOrUnknownFault" />
      <xs:enumeration value="other" />
    </xs:restriction>
  </xs:simpleType>
  <xs:complexType name="MultilingualString">
    <xs:sequence>
      <xs:element name="values">
        <xs:complexType>
          <xs:sequence>
            <xs:element name="value" type="D2LogicalModel:MultilingualStringValue"␣
↪maxOccurs="unbounded" />
          </xs:sequence>
        </xs:complexType>
      </xs:element>
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="MultilingualStringValue">
    <xs:simpleContent>
      <xs:extension base="D2LogicalModel:MultilingualStringValueType">
        <xs:attribute name="lang" type="xs:language" />
      </xs:extension>
    </xs:simpleContent>
  </xs:complexType>
  <xs:simpleType name="MultilingualStringValueType">
    <xs:restriction base="xs:string">
      <xs:maxLength value="1024" />
    </xs:restriction>
  </xs:simpleType>
  <xs:simpleType name="NonNegativeInteger">
    <xs:restriction base="xs:nonNegativeInteger" />
  </xs:simpleType>
  <xs:complexType name="NonRoadEventInformation" abstract="true">
    <xs:complexContent>
      <xs:extension base="D2LogicalModel:SituationRecord">
        <xs:sequence>
          <xs:element name="nonRoadEventInformationExtension" type=
↪"D2LogicalModel:_ExtensionType" minOccurs="0" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="OccupancyChangeValue">
```

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```
    <xs:complexContent>
      <xs:extension base="D2LogicalModel:DataValue">
        <xs:sequence>
          <xs:element name="occupancyChange" type="D2LogicalModel:Integer"␣
↪minOccurs="1" maxOccurs="1" />
          <xs:element name="occupancyChangeValueExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="OverallPeriod">
    <xs:sequence>
      <xs:element name="overallStartTime" type="D2LogicalModel:DateTime" minOccurs=
↪"1" maxOccurs="1" />
      <xs:element name="overallEndTime" type="D2LogicalModel:DateTime" minOccurs="0
↪" maxOccurs="1" />
      <xs:element name="overallPeriodExtension" type="D2LogicalModel:_ExtensionType
↪" minOccurs="0" />
    </xs:sequence>
  </xs:complexType>
  <xs:simpleType name="PassengerCarUnitsPerHour">
    <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
  </xs:simpleType>
  <xs:complexType name="PayloadPublication" abstract="true">
    <xs:sequence>
      <xs:element name="feedType" type="D2LogicalModel:String" minOccurs="0"␣
↪maxOccurs="1" />
      <xs:element name="publicationTime" type="D2LogicalModel:DateTime" minOccurs=
↪"1" maxOccurs="1" />
      <xs:element name="publicationCreator" type=
↪"D2LogicalModel:InternationalIdentifier" />
      <xs:element name="payloadPublicationExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
    </xs:sequence>
    <xs:attribute name="lang" type="D2LogicalModel:Language" use="required" />
  </xs:complexType>
  <xs:complexType name="PcuFlowValue">
    <xs:complexContent>
      <xs:extension base="D2LogicalModel:DataValue">
        <xs:sequence>
          <xs:element name="pcuFlowRate" type=
↪"D2LogicalModel:PassengerCarUnitsPerHour" minOccurs="1" maxOccurs="1" />
          <xs:element name="pcuFlowValueExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:simpleType name="Percentage">
    <xs:restriction base="D2LogicalModel:Float" />
  </xs:simpleType>
  <xs:simpleType name="ProbabilityOfOccurrenceEnum">
    <xs:restriction base="xs:string">
      <xs:enumeration value="certain" />
      <xs:enumeration value="probable" />
      <xs:enumeration value="riskOf" />
    </xs:restriction>
  </xs:simpleType>
  <xs:complexType name="Situation">
    <xs:sequence>
```

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```
      <xs:element name="situationVersionTime" type="D2LogicalModel:DateTime"␣
↪minOccurs="0" maxOccurs="1" />
      <xs:element name="headerInformation" type="D2LogicalModel:HeaderInformation"␣
↪/>
      <xs:element name="situationRecord" type="D2LogicalModel:SituationRecord"␣
↪maxOccurs="unbounded" />
      <xs:element name="situationExtension" type="D2LogicalModel:_ExtensionType"␣
↪minOccurs="0" />
    </xs:sequence>
    <xs:attribute name="id" type="xs:string" use="required" />
    <xs:attribute name="version" type="xs:string" use="required" />
  </xs:complexType>
  <xs:complexType name="SituationPublication">
    <xs:complexContent>
      <xs:extension base="D2LogicalModel:PayloadPublication">
        <xs:sequence>
          <xs:element name="situation" type="D2LogicalModel:Situation" minOccurs="0
↪" maxOccurs="unbounded" />
          <xs:element name="situationPublicationExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="SituationRecord" abstract="true">
    <xs:sequence>
      <xs:element name="situationRecordCreationTime" type="D2LogicalModel:DateTime
↪" minOccurs="1" maxOccurs="1" />
      <xs:element name="situationRecordVersionTime" type="D2LogicalModel:DateTime"␣
↪minOccurs="1" maxOccurs="1" />
      <xs:element name="probabilityOfOccurrence" type=
↪"D2LogicalModel:ProbabilityOfOccurrenceEnum" minOccurs="1" maxOccurs="1" />
      <xs:element name="source" type="D2LogicalModel:Source" minOccurs="0" />
      <xs:element name="validity" type="D2LogicalModel:Validity" />
      <xs:element name="generalPublicComment" type="D2LogicalModel:Comment"␣
↪minOccurs="0" maxOccurs="unbounded" />
      <xs:element name="groupOfLocations" type="D2LogicalModel:GroupOfLocations" />
      <xs:element name="situationRecordExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
    </xs:sequence>
    <xs:attribute name="id" type="xs:string" use="required" />
    <xs:attribute name="version" type="xs:string" use="required" />
  </xs:complexType>
  <xs:complexType name="Source">
    <xs:sequence>
      <xs:element name="sourceCountry" type="D2LogicalModel:CountryEnum" minOccurs=
↪"0" maxOccurs="1" />
      <xs:element name="sourceIdentification" type="D2LogicalModel:String"␣
↪minOccurs="0" maxOccurs="1" />
      <xs:element name="sourceName" type="D2LogicalModel:MultilingualString"␣
↪minOccurs="0" maxOccurs="1" />
      <xs:element name="sourceType" type="D2LogicalModel:SourceTypeEnum" minOccurs=
↪"0" maxOccurs="1" />
      <xs:element name="reliable" type="D2LogicalModel:Boolean" minOccurs="0"␣
↪maxOccurs="1" />
      <xs:element name="sourceExtension" type="D2LogicalModel:_ExtensionType"␣
↪minOccurs="0" />
    </xs:sequence>
  </xs:complexType>
  <xs:simpleType name="SourceTypeEnum">
    <xs:restriction base="xs:string">
```

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```
      <xs:enumeration value="automobileClubPatrol" />
      <xs:enumeration value="cameraObservation" />
      <xs:enumeration value="freightVehicleOperator" />
      <xs:enumeration value="inductionLoopMonitoringStation" />
      <xs:enumeration value="infraredMonitoringStation" />
      <xs:enumeration value="microwaveMonitoringStation" />
      <xs:enumeration value="mobileTelephoneCaller" />
      <xs:enumeration value="nonPoliceEmergencyServicePatrol" />
      <xs:enumeration value="otherInformation" />
      <xs:enumeration value="otherOfficialVehicle" />
      <xs:enumeration value="policePatrol" />
      <xs:enumeration value="privateBreakdownService" />
      <xs:enumeration value="publicAndPrivateUtilities" />
      <xs:enumeration value="registeredMotoristObserver" />
      <xs:enumeration value="roadAuthorities" />
      <xs:enumeration value="roadOperatorPatrol" />
      <xs:enumeration value="roadsideTelephoneCaller" />
      <xs:enumeration value="spotterAircraft" />
      <xs:enumeration value="trafficMonitoringStation" />
      <xs:enumeration value="transitOperator" />
      <xs:enumeration value="vehicleProbeMeasurement" />
      <xs:enumeration value="videoProcessingMonitoringStation" />
    </xs:restriction>
  </xs:simpleType>
  <xs:simpleType name="String">
    <xs:restriction base="xs:string">
      <xs:maxLength value="1024" />
    </xs:restriction>
  </xs:simpleType>
  <xs:simpleType name="TrafficStatusEnum">
    <xs:restriction base="xs:string">
      <xs:enumeration value="impossible" />
      <xs:enumeration value="congested" />
      <xs:enumeration value="heavy" />
      <xs:enumeration value="freeFlow" />
      <xs:enumeration value="unknown" />
    </xs:restriction>
  </xs:simpleType>
  <xs:complexType name="TrafficStatusValue">
    <xs:complexContent>
      <xs:extension base="D2LogicalModel:DataValue">
        <xs:sequence>
          <xs:element name="trafficStatusValue" type=
↪"D2LogicalModel:TrafficStatusEnum" minOccurs="1" maxOccurs="1" />
          <xs:element name="trafficStatusValueExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:simpleType name="UrgencyEnum">
    <xs:restriction base="xs:string">
      <xs:enumeration value="extremelyUrgent" />
      <xs:enumeration value="urgent" />
      <xs:enumeration value="normalUrgency" />
    </xs:restriction>
  </xs:simpleType>
  <xs:complexType name="Validity">
    <xs:sequence>
      <xs:element name="validityStatus" type="D2LogicalModel:ValidityStatusEnum"␣
↪minOccurs="1" maxOccurs="1" />
```

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```
      <xs:element name="validityTimeSpecification" type=
↪"D2LogicalModel:OverallPeriod" />
      <xs:element name="validityExtension" type="D2LogicalModel:_ExtensionType"␣
↪minOccurs="0" />
    </xs:sequence>
  </xs:complexType>
  <xs:simpleType name="ValidityStatusEnum">
    <xs:restriction base="xs:string">
      <xs:enumeration value="active" />
      <xs:enumeration value="suspended" />
      <xs:enumeration value="definedByValidityTimeSpec" />
    </xs:restriction>
  </xs:simpleType>
  <xs:complexType name="VehicleCountValue">
    <xs:complexContent>
      <xs:extension base="D2LogicalModel:DataValue">
        <xs:sequence>
          <xs:element name="vehicleCount" type="D2LogicalModel:NonNegativeInteger"␣
↪minOccurs="1" maxOccurs="1" />
          <xs:element name="vehicleCountValueExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="VehicleFlowValue">
    <xs:complexContent>
      <xs:extension base="D2LogicalModel:DataValue">
        <xs:sequence>
          <xs:element name="vehicleFlowRate" type="D2LogicalModel:VehiclesPerHour"␣
↪minOccurs="1" maxOccurs="1" />
          <xs:element name="vehicleFlowValueExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:simpleType name="VehiclesPerHour">
    <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
  </xs:simpleType>
  <xs:complexType name="VersionedReference">
    <xs:attribute name="id" type="xs:string" use="required" />
    <xs:attribute name="version" type="xs:string" use="required" />
  </xs:complexType>
</xs:schema>
```

Bibliography

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[16157-2] CEN TS 16157-2: Intelligent transport systems - DATEX II data exchange specifications for traffic management and information - Part 2: Location referencing

[16157-3] CEN TS 16157-3: Intelligent transport systems - DATEX II data exchange specifications for traffic management and information - Part 3: Situation Publication

[16157-4] CEN TS 16157-4: Intelligent transport systems - DATEX II data exchange specifications for traffic management and information - Part 4: Variable Message Sign (VMS) Publications

[16157-5] CEN TS 16157-5: Intelligent transport systems - DATEX II data exchange specifications for traffic management and information - Part 5: Measured and elaborated data publications

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[d2supp] DATEX II Current version/Supporting, http://www.datex2.eu/current-version-supporting (available only for registered users).

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